Direct value head across three PDEs

Whole-function value RMSE on matched frozen state banks · lower is better

Method	Concave Cubic-Quadratic PDE					Convex Cubic-Quadratic PDE					Fourier–Tanh Coupled PDE				
	20	50	100	200	400	20	50	100	200	400	20	50	100	200	400
Direct value head	9.86e-4	1.18e-3	2.36e-3	3.11e-3	5.29e-3	**5.79e-4**	1.23e-3	1.97e-3	4.13e-3	7.59e-3	**7.12e-3**	**1.47e-2**	**1.18e-2**	**4.29e-2**	**5.98e-2**
DPI	**8.81e-4**	**9.67e-4**	**2.14e-3**	**2.76e-3**	6.82e-3	7.28e-4	**9.71e-4**	**1.86e-3**	**2.93e-3**	**4.09e-3**	2.06e-2	6.25e-2	6.46e-2	2.46e-1	3.90e-1
PINN-HTE	3.83e-3	2.80e-3	3.23e-3	3.43e-3	**5.26e-3**	1.62e-3	2.68e-3	3.27e-3	4.60e-3	5.06e-3	4.90e-2	6.55e-2	7.71e-2	2.62e-1	4.01e-1
SDGD-PINN	2.59e-2	1.80e-2	2.33e-2	2.79e-2	2.97e-2	1.33e-2	1.50e-2	1.28e-2	1.68e-2	1.38e-2	1.35e-2	5.13e-2	7.41e-2	2.09e-1	2.33e-1

Direct-head pairwise wins: 35/45 (77.8%) · vs DPI 7/15 · vs PINN-HTE 13/15 · vs SDGD-PINN 15/15

Bold = best · underline = second best